Electrical and Electronics Engineering and Robotics

ELEC345 High Speed Communications
Conductors at RF and Transmission Lines

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High Speed Communications Wired and Wireless

- 10-11 lectures / tutorials in Fitzroy 107 9am
- Tutorials on alternate weeks, Smeaton 307 11am
- 11 labs in Smeaton 307 2pm
- Exam
  - 70% of Module mark
  - 5 Questions, answer four
- Coursework
  - 30% of Module mark
  - A design exercise alongside several benchmark tests
  - Presented in week 12 along with a technical note summarising the work.
- Staff
  - Dr Toby Whitley. Reynolds 114
  - John Welsh. Technical office and Smeaton 307

Topics, not in order

- Conductors at RF and Transmission lines
- dBs and link budgets
- Antenna
- Cellular systems and channel access
- L Matching and Smith Charts
- The Wireless environment
- Diversity Techniques and diversity combination
- Wi-Fi
- MIMO
- Stub matching
- RF Filters

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What are transmission lines?

- The following slides and lectures give an introduction to their basic theory, behaviour and properties.
- Why are they important?
  - A means of transmitting energy with reduced loss over a conductor
  - A way of maintaining the integrity of data and signals
- When does a wire become a transmission line?
  - Frequency, wavelength, power, length, diameter??
- What are their practical applications and limitations?
  - Connecting our modern world and its devices.
- Examples of common transmission lines will be given

Why are they needed?

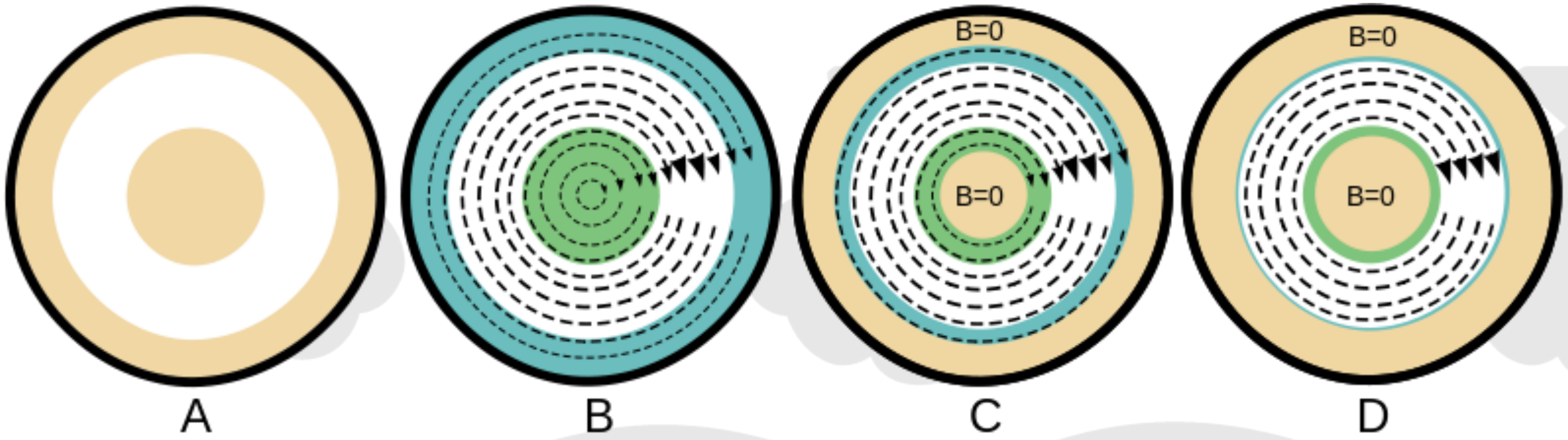
- Wires at DC
  - Resistance is set by:
    - Material
    - Dimensions
  - Resistance = Length/(Conductivity * Cross Sectional Area(CSA))
  - Resistance = Length * Resistivity / CSA
  - At DC the whole conductor is used to carry the current. Double the CSA half the resistance (ohms law)

- AC resistance
  - Increases with frequency
  - Why?

Skin Effect

- AC and more so at RF frequencies
  - Inductance
  - Increases with frequency
    - $X_L = jwL$ where $w = 2\pi f$ so $X_L = j\,2\pi f\,L$
  - Caused by magnetic flux and the density of the field line through the conductor.
  - As frequency increases so does the density of the magnetic flux lines in the conductor (induced inductance).
  - This increases the AC resistance but not in a uniform way.
  - The result is the RF current flow migrating to the outer edges of the conductor to flow along the path of least resistance.

Skin effect



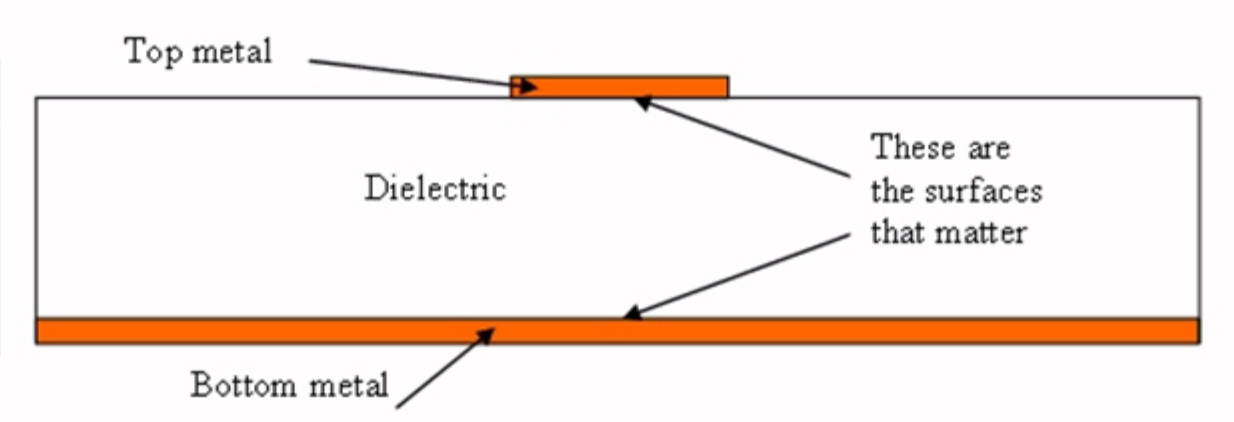
A: No current flow

B: Low frequency current flow

C: Medium Frequency

D: High Frequency

Skin effect



Where will the currents be flowing?

High frequency in micro strip. Take note of the field diagram of this later in the slides.

Skin Depth

- These skin effects lead to a reduction in the effective RF cross section of the conductor.
- Skin depth is defined as the depth at which the current density has fallen by 1/e or 37% of its surface value.

- **Skin Depth Formula**
  - $$\delta_s = \sqrt{\left(\frac{1}{\pi f \mu \sigma}\right)} = \sqrt{\frac{2}{\omega \mu \sigma}} = \sqrt{\frac{\rho}{\pi f \mu}}$$

  $\omega = 2\pi f$
- Where f is the operating frequency (Hz)
- μ is the permeability of the conductor (H/m)
  - $\mu = \mu_r \mu_0$ (μ_r is usually 1 and $\mu_0 = 4\pi * 10^{-7}$)
- σ is the conductivity of the conductor (S/m)
- ρ is resistivity (Ω*m)
  - » Dependant on only three things

  - (Bulk resistivity is a measure of how resistive a material is. It is the reciprocal of bulk conductivity.)

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$$\delta_{Gold} = \sqrt{\frac{2.44\times10^{-8}}{\pi\times1\times10^{6}\times4\pi\times10^{-7}}} =$$

Calculations

- Skin depths of common materials at various frequencies:

ρ (μΩ/cm) (Ω*10^{-8}/m)	Material	Frequency	Skin depth (μm)
1.673	Copper	1MHz	**66**
	Copper	500MHz	**2.9**
	Copper		
1.59	Silver	1MHz	**64**
	Silver	500MHz	**2.8**
	Silver		
2.44	Gold	1MHz	
	Gold	500MHz	
	Gold		

Q? What would be the skin depth of the perfect conductor?

$\mu = \mu_r \mu_0$ (μ_r is usually 1 and $\mu_0 = 4\pi*10^{-7}$)

$$\delta_s = \sqrt{\left(\frac{1}{\pi f \mu \sigma}\right)} = \sqrt{\frac{2}{\omega\mu\sigma}} = \sqrt{\frac{\rho}{\pi f \mu}}$$

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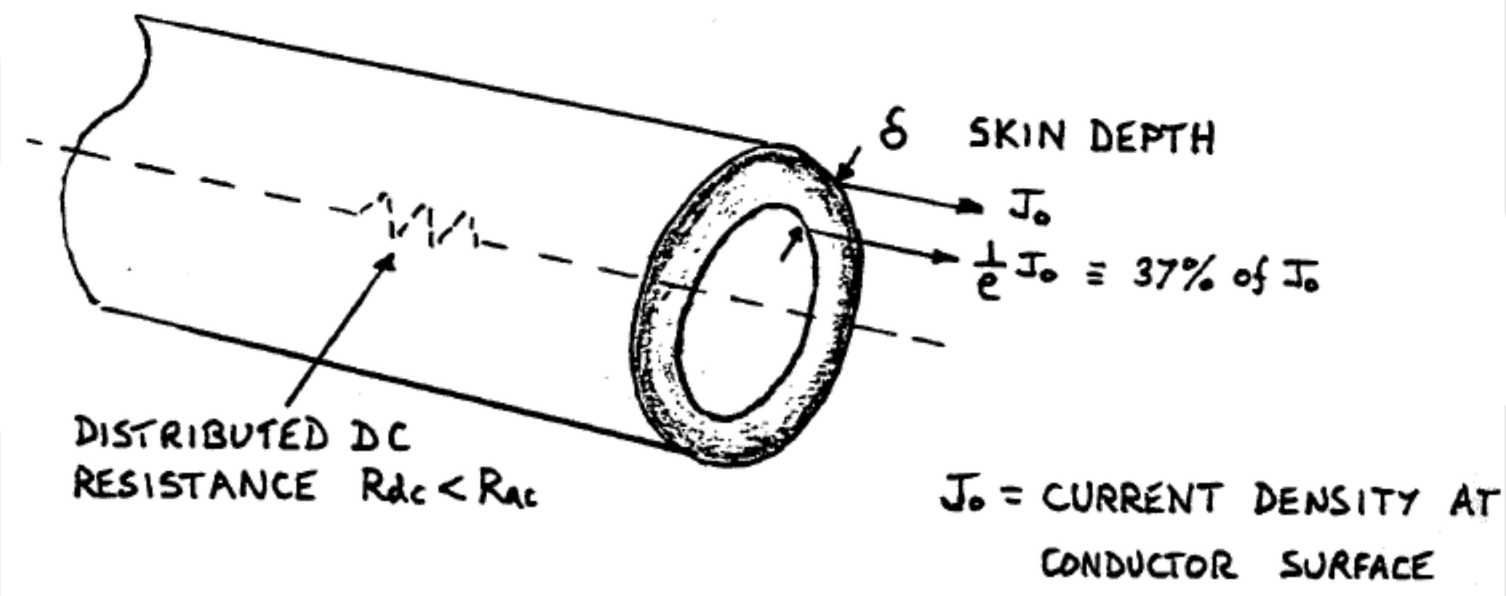
Skin Depth, implications



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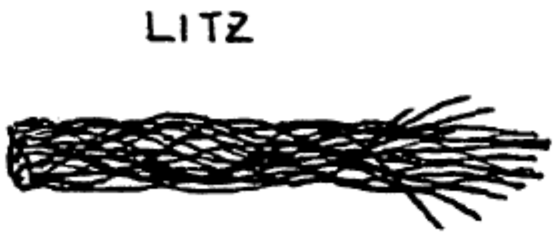
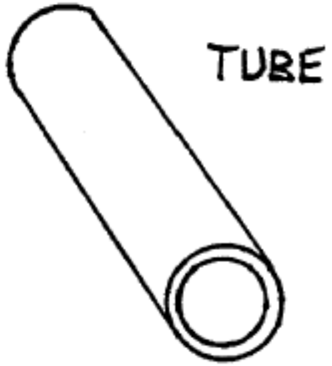
Skin Depth, implications





How thick do things need to be?
What implications does this have in terms of design and Construction?

Self inductance of conductors

- It is clear from looking at skin depth effect and what we know of transmission line properties and models that conductors have distributed inductance associated with them.
- The self inductance of connecting wire is approximately **1nH/mm** which is useful to know.
- A better estimate for a circular section conductor may be obtained using the following expression:

$$L = 0.02l\left[2.3\log\left(\frac{4l}{d} - 0.75\right)\right]\,[\mu H]$$

- **Question:** If the 1nH/mm is accurate what is the diameter of the connecting wire? If the wire diameter is increased to 3mm what effect does this have?
- **Where l = wire length in cm, d = diameter of wire in cm, L = self inductance**

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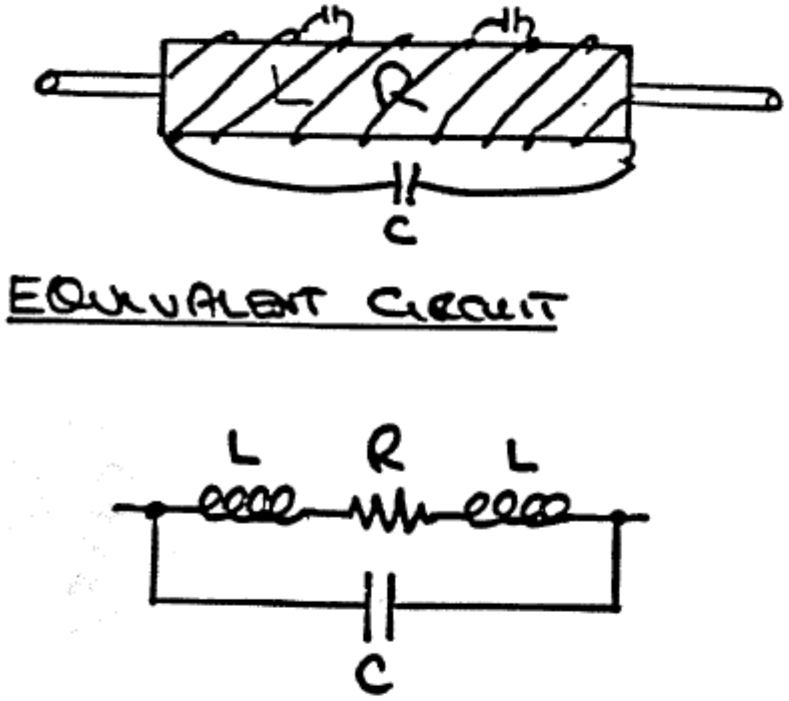
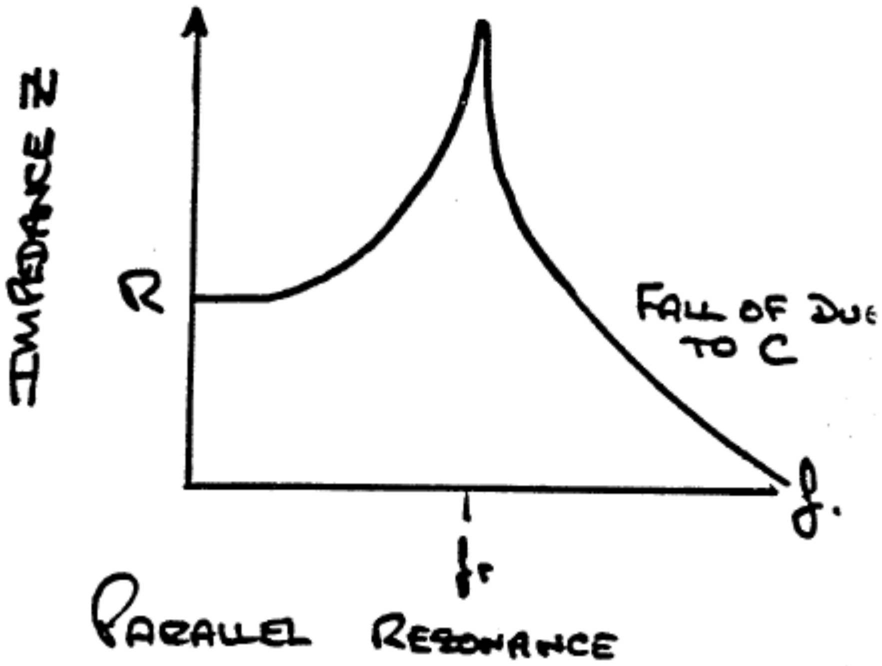
Lumped components : Resistors

Wire Wound Resistors

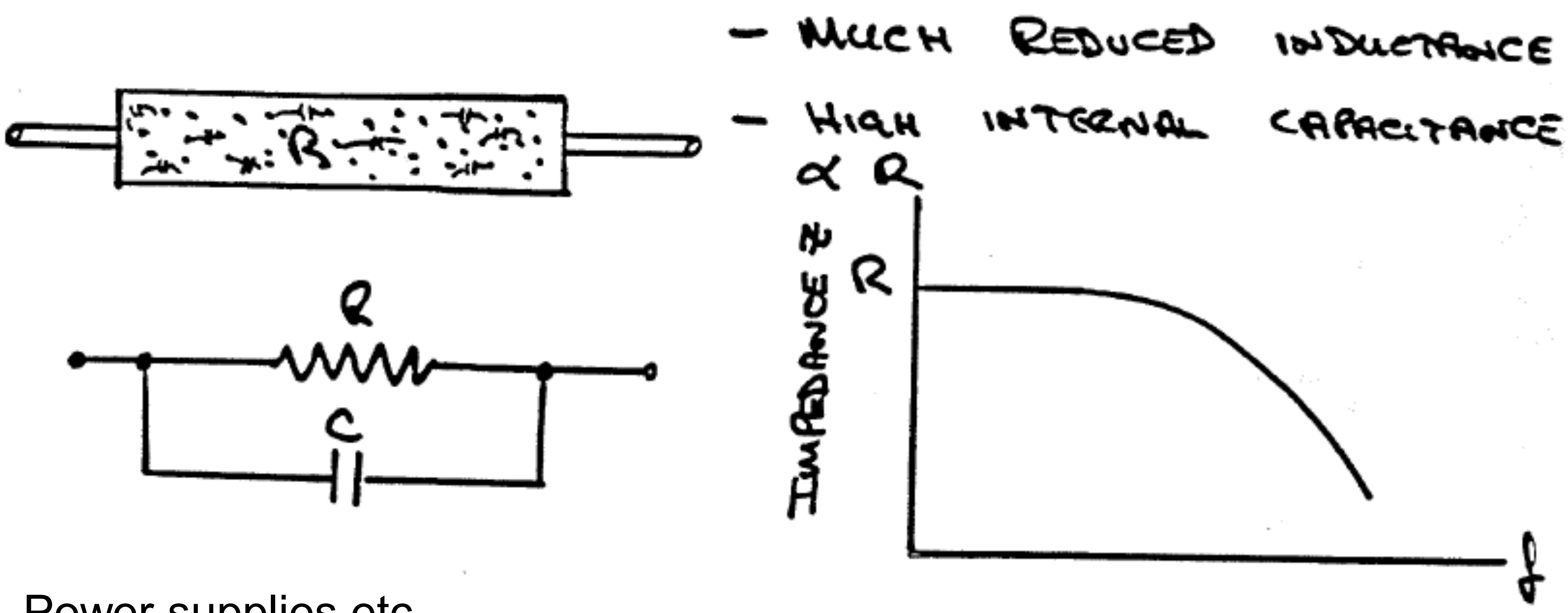


- HIGHLY INDUCTIVE ∝ R
- COIL INDUCTANCE SWAMPS LEAD INDUCTANCE



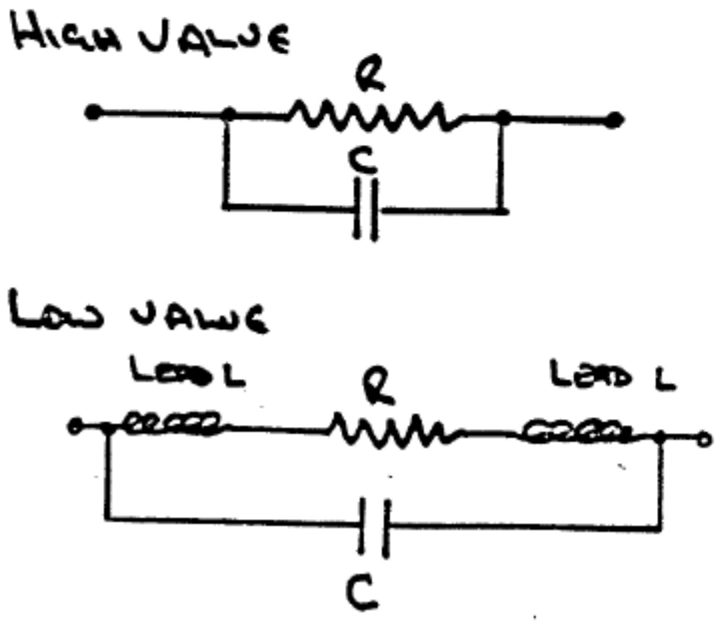
Resistors

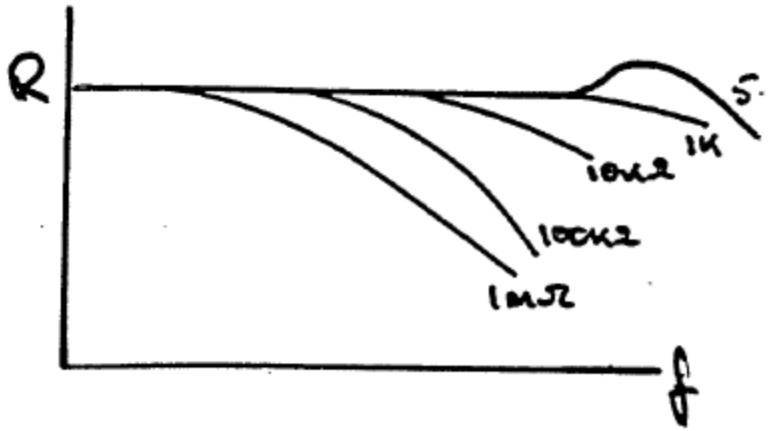
Carbon Composition



Power supplies etc.
Expensive

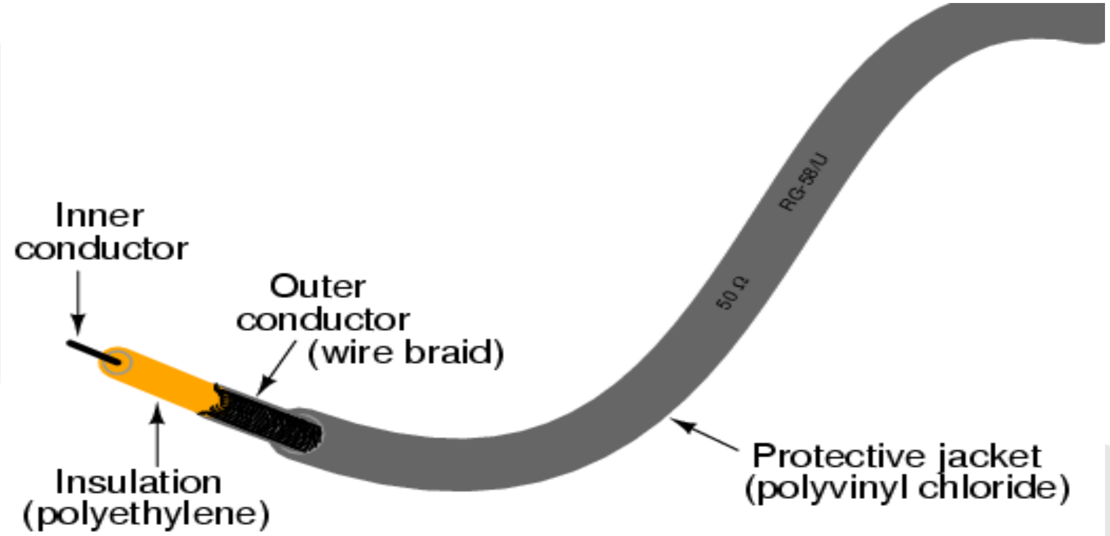
Resistors





RF and microwave applications

What does make a transmission line?

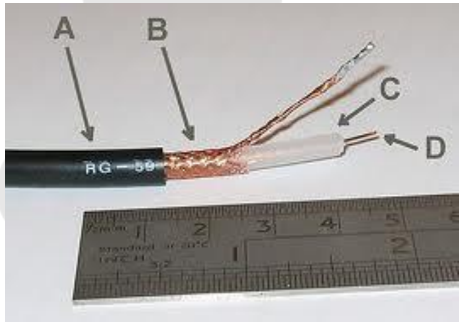


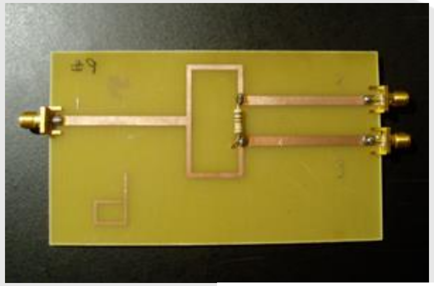
- Physical properties:
  - Two (parallel) conductors with either a signal and ground (un-balanced) or the same signal referenced to ground in both, but one is 180 degrees out of phase (balanced).

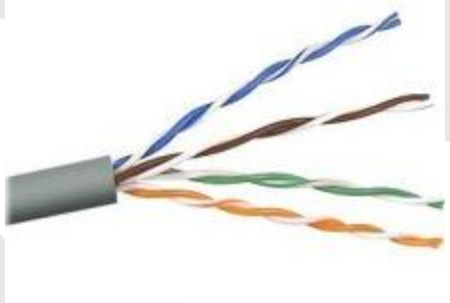
Other commonly seen transmission lines.

- Coaxial cable
- TV and RF cables
- Stripline
- Microstrip

    Cheaper and more widely used (PCB design)
- Twisted pair (balanced lines)
    - Widely used in everything from telephones to LANs







In transmission lines, size really does matter.

- The length of the cable.

  >0.1λ, one tenth of the wavelength of the transmitted signal.
- The frequency or wavelength of the transmitted signal.

  Circuit theory is insufficient
- Ordinary cabling is not suitable for carrying these signals efficiently.
  - Reflection
  - Radiation
    - Equals losses

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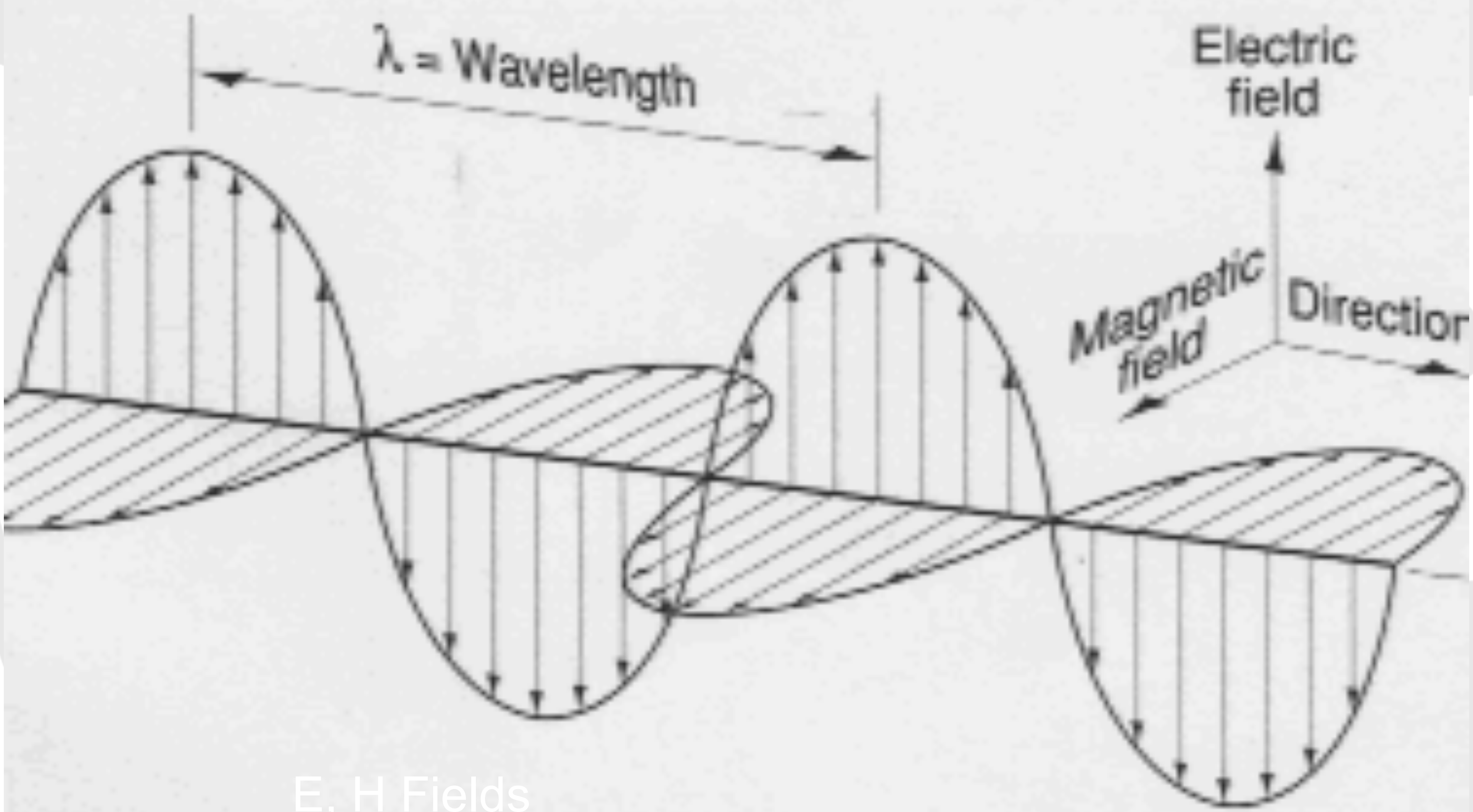
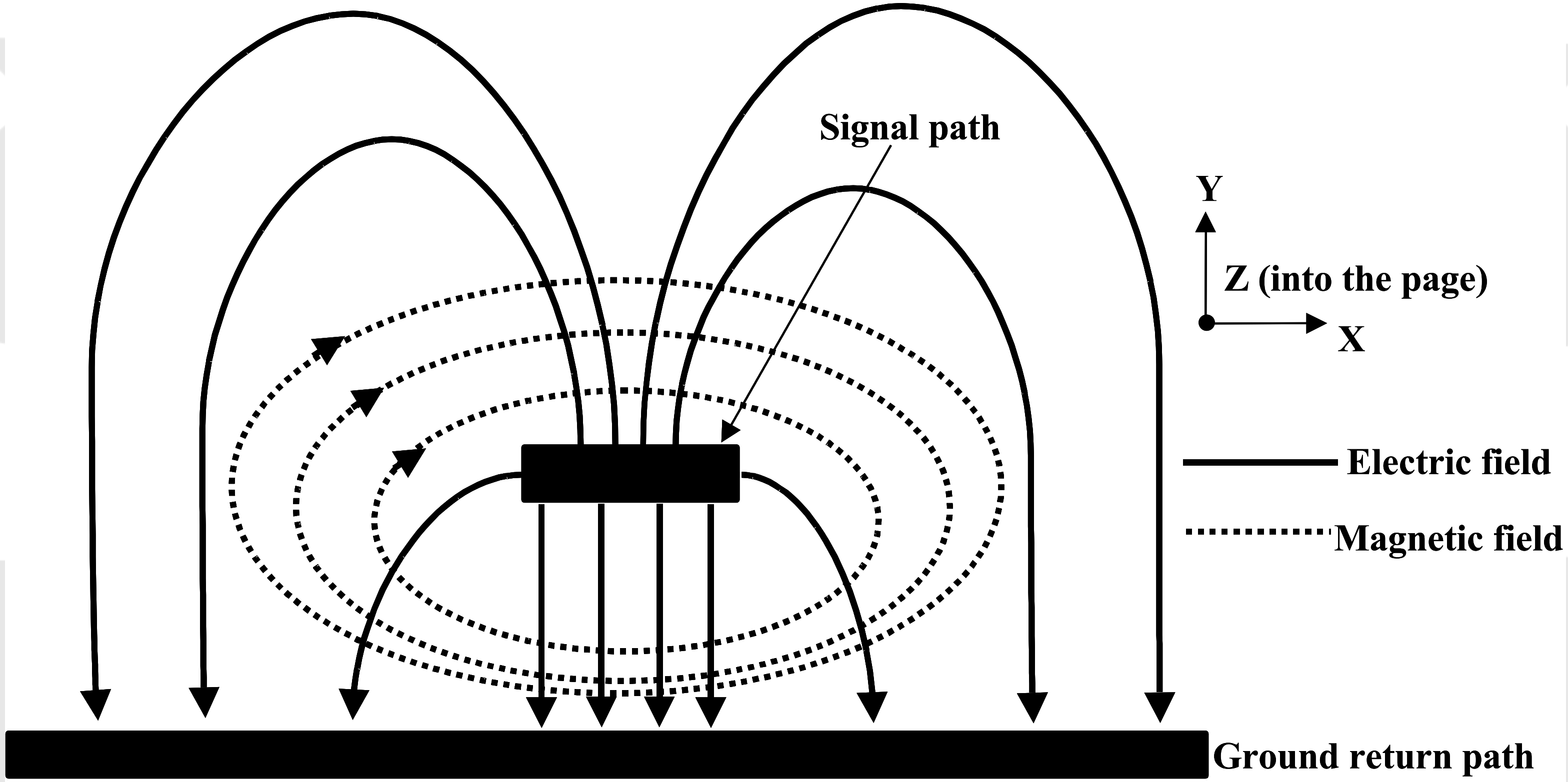


E, H Fields

E, H Fields for micro strip.



The signal is really the wave propagating between the conductors

A sense of scale

- The speed of light in free space is a top speed, but, in transmission lines and on copper this speed is considerably reduced.

- Free space, c = 300000000 m/s
- Transmission line, c = ~240000000 m/s

Frequency (GHz)	Wavelength λ (cm)	0.1λ (cm)
0.1	300	30
1	30	3
1.5	20	2
2	15	1.5
2.5	12	1.2
3	10	1
5	6	0.6
10	3	0.3
100	0.3	0.03

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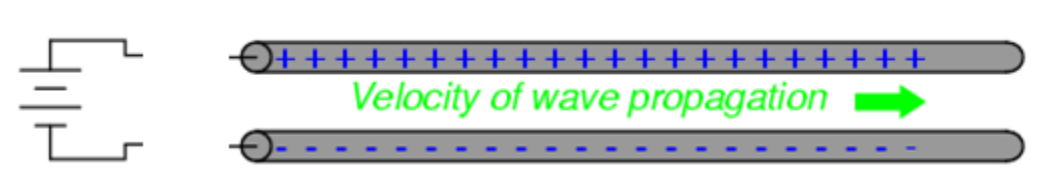
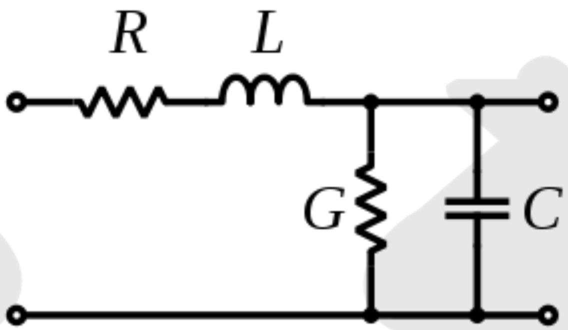
Characteristic impedance

- IMAGINE:
- A transmission line.
- Of infinite length

- With a battery disconnected.
- What happens when the battery is connected?



Transmission line models



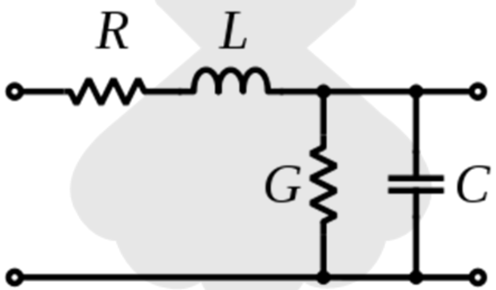
Resistance (R), Inductance (L), Conductance (G), Capacitance (C), all per unit length.

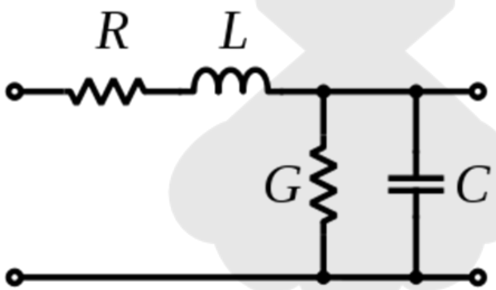
Telegraphers Equations:

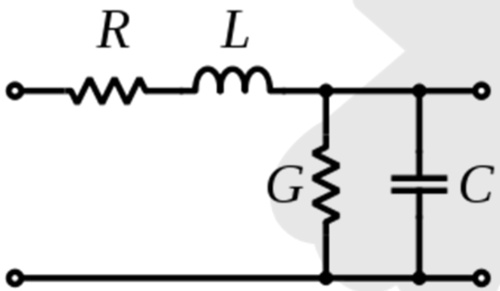
- Linear differential equations that allow the voltage and current to be found with respect to distance and time along a transmission line.
- A key expression is:

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}.$$

- Which gives the Characteristic Impedance

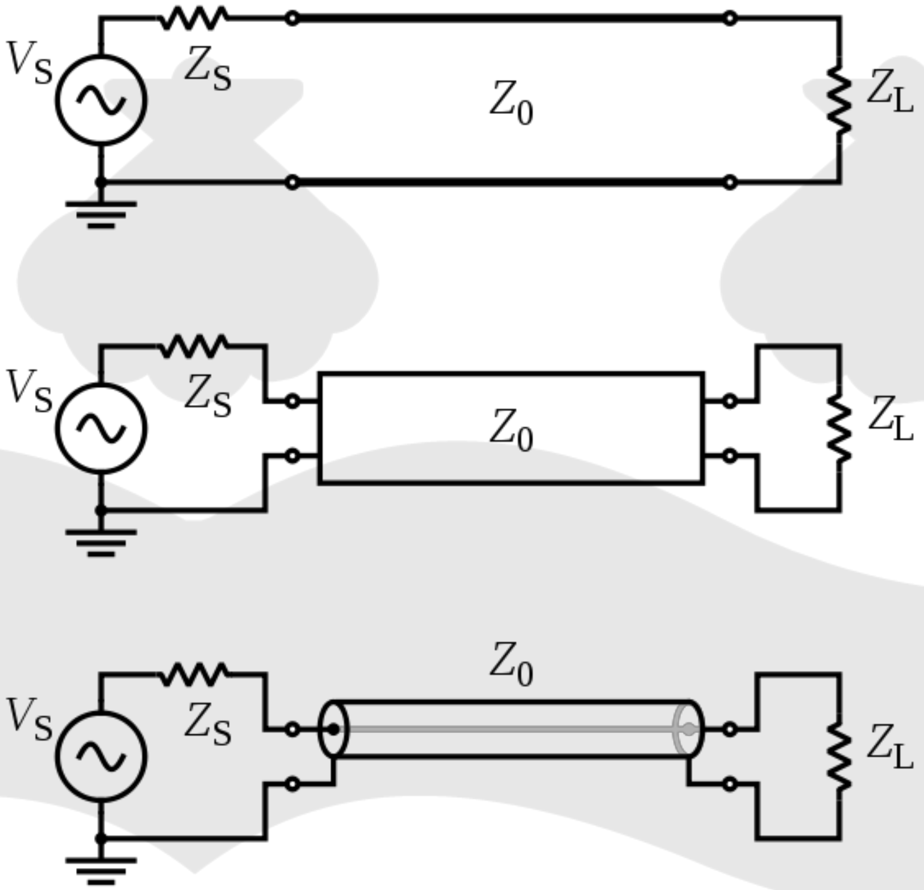






Transmission line model

Transmission line models



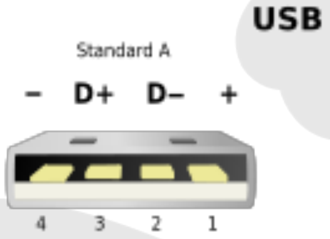
- Where:
- Zs = input impedance
- Z0 = characteristic impedance
- ZL = load impedance

- Special case:
  - The matched load

The universal serial bus (USB)

- USB 1.1 (~1998)
- 1.5 Mbit/s Low speed (Max 3m)
- 12 Mbit/s Full speed (Max 5m)
- USB 2 (~2000)
- 480 Mbit/s High speed(Max 5m)
- USB 3 (~2010)
- 5 Gbit/s Super speed(Max ~??)

- How has this been achieved?



Twisted pair to reduce noise and crosstalk. Characteristic impedance of ~90Ω

Serial advanced technology attachment (SATA)

A bus interface standard for connecting data storage devices like hard drives and optical drives to a host computer bus adaptor. Connects inside a PC

Pin	Function
1	Ground
2	A+ (transmit)
3	A- (transmit)
4	Ground
5	B− (receive)
6	B+ (receive)
7	Ground
-	Coding Notch

•Replaced parallel ATA and now provides data rates of 1.5, 3 and now 6 Gbit/s
•Allows longer cables, 1m compared to 40cm
•Smaller cables allow better airflow in chassis

Why change from parallel ATA (IDE).

- A better transmission line for the job
- Higher speed
- Better noise performance

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Peripheral component interconnect express (PCIe)

- Replaces older bus standards
- PCI Express 1x250 [500]* MB/s
- PCI Express 2x500 [1000]* MB/s
- PCI Express 4x1000 [2000]* MB/s
- PCI Express 8x2000 [4000]* MB/s
- PCI Express 16x4000 [8000]* MB/s
- PCI Express 32x8000

Gigabit Ethernet

- Wired LAN standard.
  - How has it evolved?
  - Network protocols, packet size.
  - Cabling (The transmission lines)
- 1000BASE-T Twisted-pair cabling (Cat-5, Cat-5e, Cat-6, or Cat-7)100 meters
- 1000BASE-TX Twisted-pair cabling (Cat-6, Cat-7)100 meters

Pair	Wire	Color
3	tip	white/green
3	ring	green
2	tip	white/orange
1	ring	blue
1	tip	white/blue
2	ring	orange
4	tip	white/brown
4	ring	brown

Summary

Just a wire?

- Physical and electrical properties of transmission lines
  - Physical, the properties and design issues of cables.
  - Electrical, the circuit characteristics and equivalent models that are important in the specification and design of data transmission lines. How the physical properties relate to this.
  - The key theoretical and practical considerations in the use of transmission lines
- The properties and performance of a number of modern wired communication standards, USB, Gigabit Ethernet etc.
- How transmission lines are fundamental to their performance
- What happens next?

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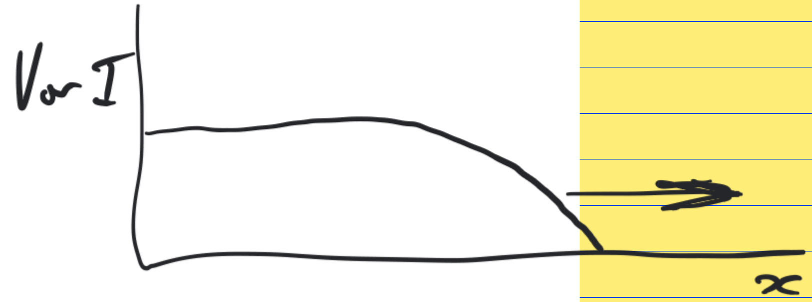
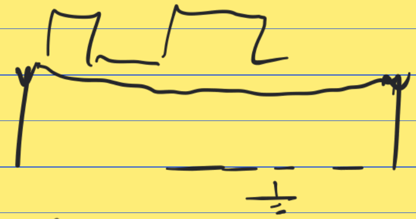
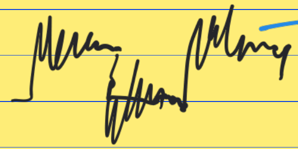
The Telegraphers equations.

Morse – 1837
Oliver Heaviside – 1880







Ringing.

Motherboard design & logic issue

how long does a pulse need to be to settle.

ISI.

To match ringing = dielec loss.

$$\Delta V = -L\Delta x \frac{\partial I}{\partial t} \qquad \Delta I = -C\Delta x \frac{\partial V}{\partial t}$$

$\div x$

$$\frac{\partial V}{\partial x} = -L\frac{\partial I}{\partial t} \qquad \frac{\partial I}{\partial x} = -C\frac{\partial V}{\partial t}$$

Variation in voltage is related to distance, time & current. vice versa for C & volt.

wrt x

$$\frac{\partial^2 V}{\partial x^2} = -L\frac{\partial^2 I}{\partial x \partial t} \qquad \frac{\partial^2 I}{\partial x^2} = -C\frac{\partial^2 V}{\partial x \partial t}$$

wrt t

$$\frac{\partial^2 V}{\partial t \partial x} = -L\frac{\partial^2 I}{\partial t^2} \qquad \frac{\partial^2 I}{\partial t \partial x} = -C\frac{\partial^2 V}{\partial t^2}$$

A bit of maths to give us V or I in terms of distance & time + medium/structure.

$$\frac{\partial^2 V}{\partial x^2} = LC\frac{\partial^2 V}{\partial t^2} \qquad \frac{\partial^2 I}{\partial x^2} = LC\frac{\partial^2 I}{\partial t^2}$$

$$\frac{\partial^2 V}{\partial x^2} - LC\frac{\partial^2 V}{\partial t^2} = 0 \qquad \frac{\partial^2 I}{\partial x^2} - LC\frac{\partial^2 I}{\partial t^2} = 0$$

$$\frac{\partial^2 A}{\partial x^2} - \frac{1}{V_p^2}\frac{\partial^2 A}{\partial t^2} = 0 \quad \text{— Wave Equation}$$

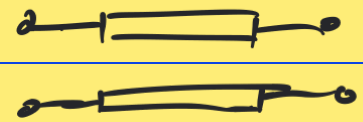
General equation for any travelling wave. sound, light, etc

$$\frac{\partial^2 V}{\partial x^2} - LC\frac{\partial^2 V}{\partial t^2} = 0 \quad \therefore \quad \frac{1}{V_p^2} = LC \quad V_p = \frac{1}{\sqrt{LC}}$$

$$ms^{-1} = \frac{1}{\sqrt{Hm^{-1}\,Fm^{-1}}}$$

A level physics. Unit check.

$$A_y = \quad y = t - \frac{x}{v} \quad \longrightarrow \qquad y = t + \frac{x}{v} \quad \longleftarrow$$



General solution, both directions. $\longrightarrow$ $\longleftarrow$

$y = t-$ $\quad y = t+$

Like

$V = V \sin \omega t.$

$V' = V\left(t - \frac{x}{v}\right)$ & $V = V\left(t + \frac{x}{v}\right)$

$\frac{dV}{dt} = \ldots \quad \frac{dV}{dx} = \ldots$

$\frac{d^2V}{dt^2} = \ldots \quad \frac{d^2V}{dx^2} = \ldots$

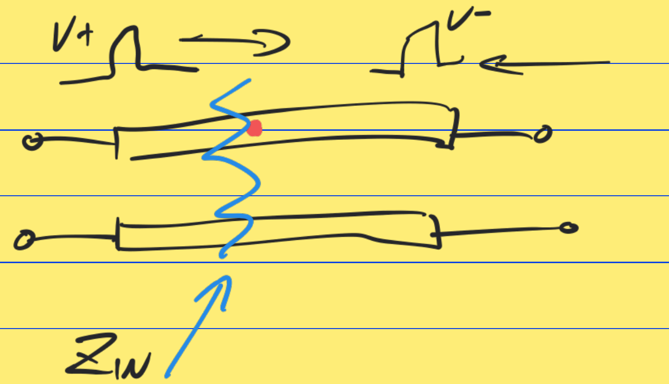
You can test this if you can!

You will go down in history and be mentioned in future lectures for years to come

V^+ & V^- are amplitudes

V^+ going left to right
V^- going right to left

Putting in our solution



$$V_{(x,t)} = \underbrace{V^+\left(t - \frac{x}{v}\right)}_{\text{fwd}} + \underbrace{V^-\left(t + \frac{x}{v}\right)}_{\text{bwd}}$$

Look at this pt on the line

$$I_{(x,t)} = \frac{V^+}{Z_0}\left(t - \frac{x}{v}\right) - \frac{V^-}{Z_0}\left(t + \frac{x}{v}\right)$$

$$R = Z_{IN} = \frac{V}{I} = Z_0 \frac{(V^+ + V^-)}{(V^+ - V^-)}$$

$Z_0 = 75\Omega$ vs 50Ω ??

$$Z_{IN} = Z_L = \frac{V}{I} = Z_0\frac{(V^+ + V^-)}{(V^+ - V^-)} \implies \frac{V^-}{V^+} = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} \Rightarrow \Gamma$$

(Greek Gamma)

Voltage reflection coeff

Matched line $= \frac{50 - 50}{50 + 50} = 0$ $\qquad |\Gamma| \leq 1$

$-\Gamma = \frac{I^-}{I^+}$ Current reflection coefficient.

REFLECTION COEFFICIENT $\Gamma = \frac{V^-}{V^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$

Current " " $= -\Gamma$

V^+ & V^- are amplitudes

V^+ going left to right
V^- going right to left

Putting in our solution

$$V_{(x,t)} = \underbrace{V^+\left(t - \frac{x}{v}\right)}_{\text{fwd}} + \underbrace{V^-\left(t + \frac{x}{v}\right)}_{\text{bwd}}$$

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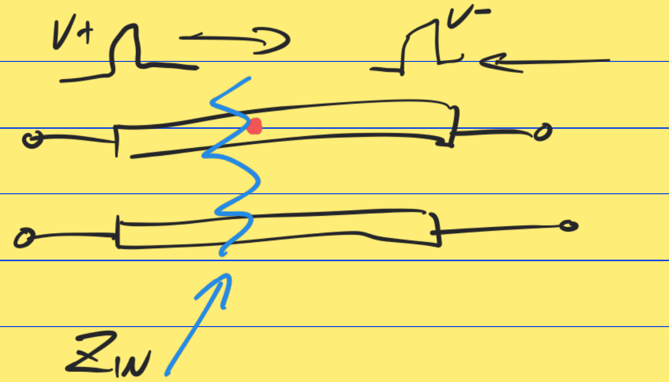


Look at this pt on the line

$$R = Z_{IN} = \frac{V}{I} = Z_0\frac{(V^+ + V^-)}{(V^+ - V^-)}$$

$Z_0 = 75\Omega$ vs 50Ω ??

$$Z_{IN} = Z_L = \frac{V}{I} = Z_0\frac{(V^+ + V^-)}{(V^+ - V^-)} \Longrightarrow \frac{V^-}{V^+} = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} \Rightarrow \Gamma$$

(Greek Gamma)

Voltage reflection coeff

Matched line $= \frac{50 - 50}{50 + 50} = 0$ $\qquad |\Gamma| \leq 1$

$$-\Gamma = \frac{I^-}{I^+}$$ Current reflection coefficient.

REFLECTION COEFFICIENT $\Gamma = \frac{V^-}{V^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$

Current " " $= -\Gamma$

Γ = reflection coefficient = $\dfrac{Z_{in} - Z_0}{Z_{in} + Z_0}$

Symetrical so $\Gamma = \dfrac{Z_L - Z_0}{Z_L + Z_0}$

What is the reflection coeff Γ =

How much gets reflected ? =

look at the animation on wikipedia

but note this is for a load less than Z_0 as it reflects/returns on the gnd.

Home work

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \qquad \Gamma = ?$$

1. Why is $Z_0 = 50\,\Omega$ chosen for RF applications & $75\,\Omega$ for terrestrial TV? (Google it)

2. What is the reflection coefficient Γ for an open circuit.

3. " short circuit

4. " matched load

so $Z_0 = Z_L = 50\,\Omega$

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

$$\Downarrow$$

Find the range of values. **Homework**

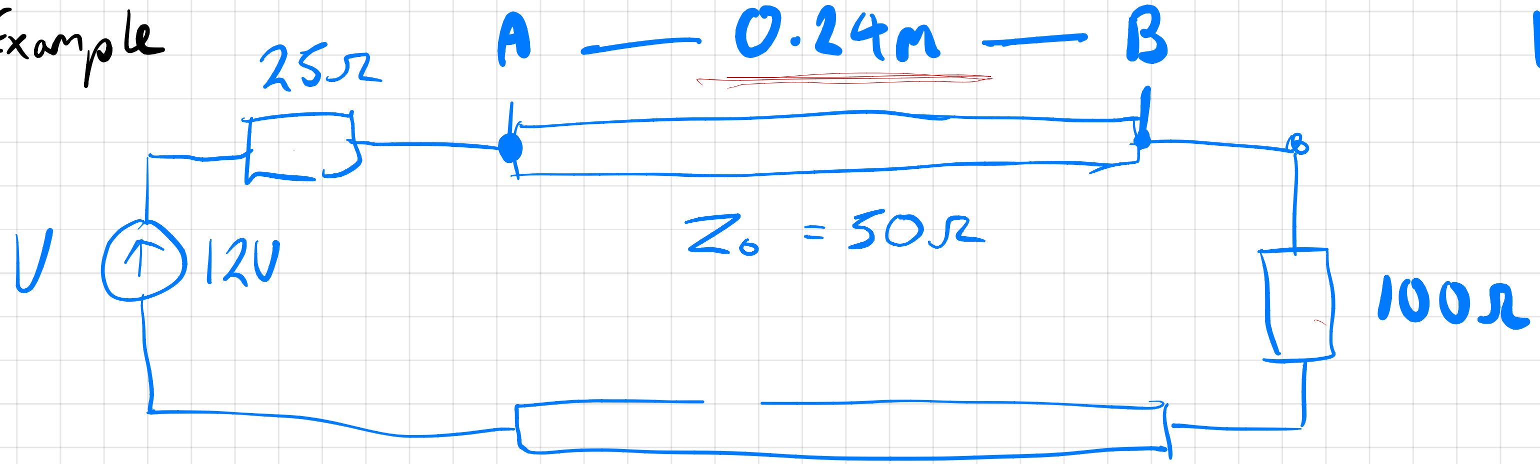
Return loss $= R_{L_{dB}} = 10 \log \frac{P_r}{P_i}$

Also $= -20 \log |\Gamma|$

Power $= V \cdot I$

$I = \frac{V}{R}$ $\quad = V \cdot \frac{V}{R}$ $\quad P = V^2$

Example



$V_p = 2.4 \times 10^8$ m/s

How much voltage goes onto the line at the start